

A Consolidated Overview of Datasets and Benchmarks for Machine Learning and Computer Vision

Tobias Schlosser 

Dresden University of Technology,
01307 Dresden, Germany,
`{firstname.lastname}@tu-dresden.de`

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Abstract

This manuscript presents a consolidated, practitioner-oriented overview of widely used datasets and benchmarks in machine learning (ML) and computer vision (CV). It provides a compact reference for selecting appropriate benchmarks, interpreting common evaluation settings, and assessing key dataset characteristics, including modality, scale, and annotation scope. In addition to canonical datasets, the manuscript covers benchmark hubs and repositories that standardize data access, partitioning, and metadata, while emphasizing the influence of dataset selection on comparability, reproducibility, and the validity of empirical conclusions. The overview spans ML and CV and encompasses representative task families, including tabular learning, time-series analysis, graph learning, recommender systems, natural language processing, reinforcement learning, and major CV domains such as classification, detection, segmentation, three-dimensional (3D) vision, remote sensing, vision-language learning, and medical imaging. Conceived as a living reference, it is intended to help newcomers and experienced researchers navigate the benchmark landscape efficiently and make transparent, well-justified dataset choices in experimental design and reporting.

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1 Introduction and Motivation

Datasets and benchmarks are fundamental to progress in machine learning (ML) and computer vision (CV): they operationalize research tasks, establish evaluation protocols, and enable quantitative comparisons among methods. Dataset selection, however, remains an important and often underappreciated source of variation in experimental outcomes. Differences in label definitions, sampling strategies, training, validation, and test partitions, annotation quality, and domain coverage can equal or exceed the effects of architectural or optimization changes. Rigorous empirical research therefore requires more than selecting a widely used dataset; it requires a clear understanding of what a benchmark measures, what it omits, and how closely its assumptions reflect the intended application or deployment context.

The benchmark landscape has expanded rapidly, from classical tabular problems to web-scale corpora for foundation models, specialized medical and scientific datasets, and increasingly sophisticated multimodal benchmarks. In parallel, data access has shifted toward standardized repositories and dataset loaders that support reproducibility through explicit versioning, integrity checks, and canonical partitions. This manuscript consolidates representative datasets and benchmark suites across ML and CV in structured tables that summarize modality, approximate scale, and principal tasks. Rather than seeking exhaustive coverage, it focuses on widely established reference points and practical starting points for experimental design, baseline construction, comparative evaluation, and transparent reporting.

Throughout this manuscript, dataset denotes a curated collection of samples—such as images, time series, text, or graphs—together with associated metadata and, where applicable, annotations. Benchmark denotes a dataset paired with a defined evaluation protocol, including metrics, partitions, constraints, and reporting conventions. The overview is designed to evolve over time and to encourage transparent documentation, including explicit reporting of dataset versions, preprocessing procedures, partitioning strategies, and evaluation protocols.

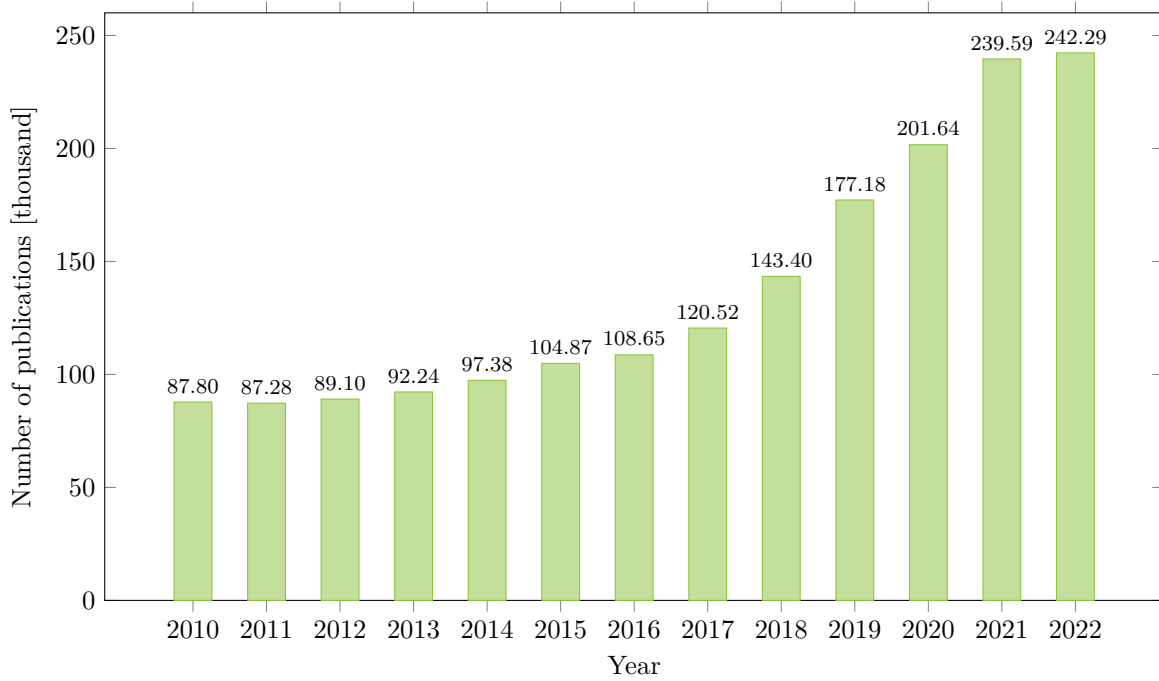


Figure 1: Number of artificial intelligence (AI) publications worldwide from 2010 to 2022 [1, p. 31].

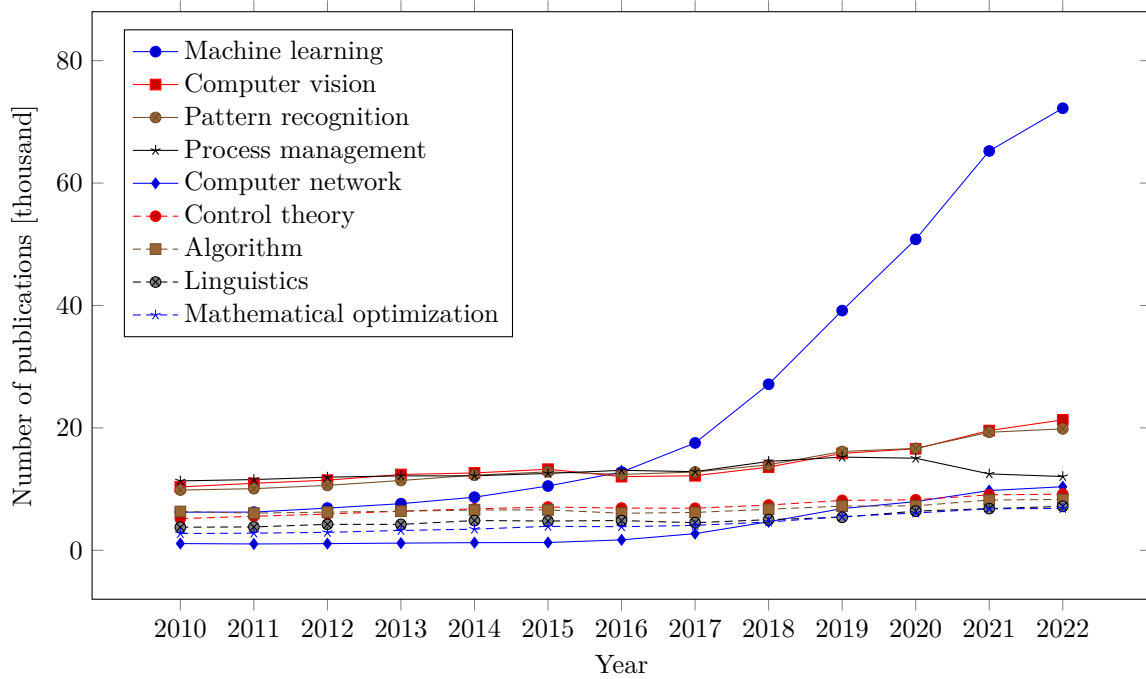


Figure 2: Number of AI publications worldwide from 2010 to 2022 by field of study [1, p. 33].

2 Machine Learning

Machine learning benchmarks encompass diverse modalities, learning paradigms, and task formulations. Classical supervised learning on tabular data remains central to structured prediction in domains such as finance, marketing, and healthcare, whereas time-series benchmarks address forecasting, classification, and anomaly-detection problems characterized by temporal dependence, nonstationarity, and seasonality. Graph benchmarks support systematic evaluation of relational reasoning and representation learning on network-structured data, while recommender-system datasets expose challenges such as sparsity, popularity bias, implicit feedback, and temporal drift. In natural language processing (NLP), benchmark suites for language understanding and question answering (QA) coexist with large pretraining corpora used to develop large language models (LLMs), making it essential to distinguish evaluation resources from pretraining data. Reinforcement learning (RL) benchmarks commonly comprise interactive environment suites or logged trajectories, and their interpretation depends strongly on the interaction protocol, particularly whether learning is online or offline.

Across ML benchmarks, data scale, annotation quality, representativeness, and evaluation stability are closely interrelated. Small, carefully curated datasets facilitate controlled experimentation but may become saturated or fail to capture the variability of realistic applications. Large-scale datasets can improve statistical power and coverage, yet they also increase exposure to dataset-specific shortcuts, spurious correlations, duplicated content, and poorly characterized distribution shifts. Some benchmarks now serve primarily as historical reference points that preserve continuity with earlier work, whereas others are designed explicitly to assess robustness, fairness, or out-of-distribution generalization. Transparent reporting should therefore identify dataset versions and canonical partitions, describe checks for leakage or overlap, and document all filtering, preprocessing, and deduplication procedures, particularly when datasets are obtained through evolving repositories.

For concision, the accompanying table uses the following abbreviations:

- 1D = one-dimensional
- ASR = automatic speech recognition
- EHR = electronic health record
- GNN = graph neural network
- HEP = high-energy physics
- ICU = intensive care unit
- IE = information extraction
- IR = information retrieval
- LLM = large language model
- MedImg = medical imaging
- MRI = magnetic resonance imaging
- MT = machine translation
- NER = named-entity recognition
- NLI = natural language inference
- NLP = natural language processing
- QA = question answering
- RecSys = recommender system
- RL = reinforcement learning
- TSC = time-series classification

Table 1: Machine learning datasets.

Dataset	Area	Modality	Scale	Typical tasks / notes
2.1 Benchmark hubs and repositories				
Hugging Face Datasets [2]	NLP/Multimodal	Mixed	1000s	Unified dataset hosting/loader; many tasks and modalities
Kaggle Datasets [3]	Multi-domain	Mixed	100k+	Community datasets; quality varies; good for rapid prototyping
OpenML [4]	Tabular	Mixed	10k+ tasks	Benchmarking hub with standardized tasks, splits, and metadata
TensorFlow Datasets (TFDS) [5]	Multimodal	Mixed	100s	Curated dataset loaders with versioning and checksums
2.2 Tabular ML (classical supervised learning)				
Adult (Census Income) [6]	Tabular	Tabular	48k	Binary classification; fairness discussions common
Bank Marketing [7]	Tabular	Tabular	45k	Marketing response prediction; imbalanced classification
Credit Card Fraud (EU) [8]	Tabular	Tabular	285k	Highly imbalanced anomaly/fraud detection
HIGGS [9]	Tabular	Tabular	11M	Large-scale binary classification; HEP-inspired
UCI ML Repository [10]	Tabular	Tabular	100s	Classic ML datasets; heterogeneous difficulty and documentation
2.3 Time series and forecasting				
ETTh/ETTm [11]	Forecast	Time series	2 years	Transformer time series benchmarks (ETT family)
M4 [12]	Forecast	Time series	100k series	Large forecasting competition dataset
M5 [13]	Forecast	Time series	Walmart	Retail demand forecasting; hierarchical aggregation
METR-LA [14]	Forecast	Traffic	207 sensors	Traffic-speed forecasting on a road-sensor network
UCR Time Series Classification Archive [15]	Time series	1D	100+ sets	Univariate time series classification benchmarks

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Dataset	Area	Modality	Scale	Typical tasks / notes
UEA Multivariate TSC Archive (2018) [16]	Time series	Multivariate	30+ sets	Multivariate time series classification benchmarks
2.4 Anomaly detection (general)				
Numenta Anomaly Benchmark (NAB) [17]	Anomaly	Time series	58 files	Streaming anomaly detection with labeled windows
UNSW-NB15 [18]	Anomaly	Tabular	2.5M	Modern intrusion detection benchmark alternative
2.5 Graph machine learning				
CiteSeer [19]	Graphs	Citation graph	3.3k nodes	Node classification; classic benchmark
Cora [20]	Graphs	Citation graph	2.7k nodes	Node classification; classic GNN benchmark
Open Graph Benchmark (OGB) [21]	Graphs	Graph	Many	Standardized graph benchmarks and evaluation protocols
QM9 [22]	Graphs	Molecules	134k	Molecular property prediction with quantum-chemical targets
Reddit (Graph) [23]	Graphs	Graph	232k nodes	Inductive node classification; community graphs
ZINC [24]	Graphs	Molecules	Large	Molecular representation learning and virtual screening
2.6 Recommender systems				
Amazon Reviews [25]	RecSys/NLP	Text+ratings	Large	Review-based recommendation; domain shift across categories
MovieLens [26]	RecSys	Ratings	Various	Collaborative filtering; multiple sizes/versions
Yelp Reviews [27]	RecSys/NLP	Text+ratings	Large	Review classification and sentiment modeling
2.7 NLP benchmarks (understanding, QA, IE)				
GLUE [28]	NLP	Text	9 tasks	General language understanding evaluation suite
HotpotQA [29]	NLP	Text	113k	Multi-hop question answering with supporting facts

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Dataset	Area	Modality	Scale	Typical tasks / notes
MS MARCO [30]	IR/QA	Text	Large	Passage ranking, retrieval, QA-style supervision
MultiNLI [31]	NLP	Text	433k	Natural language inference across multiple genres
Natural Questions [32]	NLP	Text	300k+	Open-domain question answering from real search queries
OntoNotes 5.0 [33]	NLP	Text	Large	NER, coreference; richer annotation
SQuAD [34]	NLP	Text	100k+ pairs	QA
SuperGLUE [35]	NLP	Text	8 tasks	Reading comprehension / extractive QA
TyDi QA [36]	NLP	Text	11 languages	Harder successor to GLUE
XNLI [37]	NLP	Text	15 languages	Multilingual QA across typologically diverse languages
2.8 NLP pretraining corpora				
BookCorpus [38]	NLP	Text	Large	Cross-lingual NLI benchmark
C4 [39]	NLP	Text	Large	Books; used in early transformer pre-training
Penn Treebank [40]	NLP	Text	4.5M words	Cleaned Common Crawl; popular for transformer pretraining
The Pile [41]	NLP	Text	800GB	Foundational annotated corpus for parsing and language modeling
WikiText-103 [42]	NLP	Text	103M tokens	Curated mixture for LLM pretraining
2.9 Machine translation and multilingual corpora				
FLORES-200 [43]	MT/NLP	Text	200 languages	Many-to-many MT evaluation benchmark
OPUS [44]	MT	Text	Large	Open parallel corpora collection; many domains/languages
2.10 Speech and audio				
AudioSet [45]	Audio	Audio	2M clips	Audio event classification with ontology-based labels
Common Voice [46]	Speech	Audio	Large	Multilingual ASR; community-recorded, noisy conditions

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Dataset	Area	Modality	Scale	Typical tasks / notes
IEMOCAP [47]	Speech	Audio/Video	12 h	Multimodal emotional speech and interaction corpus
Libri-Light [48]	Speech	Audio	60k h	Large-scale speech corpus for limited- and self-supervised ASR
LibriSpeech [49]	Speech	Audio	1000h	ASR benchmark (audiobooks)
Speech Commands [50]	Speech	Audio	105k	Limited-vocabulary keyword-spotting benchmark
VoxCeleb [51]	Speech	Audio/Video	Large	Speaker verification/identification in the wild
2.11 Reinforcement learning (environments and offline RL)				
Arcade Learning Environment (ALE) [52]	RL	Games	57 games	Atari-style RL benchmark suite
D4RL [53]	RL	Trajectories	Many	Offline RL datasets + standardized evaluation
DeepMind Control Suite [54]	RL	Simulation	Many tasks	Continuous control benchmarks
MuJoCo Benchmarks [55]	RL	Simulation	Many tasks	Locomotion/manipulation; common in policy learning
2.12 Healthcare, EHR data, and clinical prediction				
eICU Collaborative Research Database [56]	EHR	Structured	Large	Multi-center ICU EHR; diagnosis/outcome prediction
MIMIC-III [57]	EHR	Structured notes	Large	ICU EHR benchmark; many clinical prediction tasks
MIMIC-IV [58]	EHR	Structured notes	Large	Newer ICU EHR release
2.13 Medical imaging (ML-relevant benchmarks)				
BraTS [59]	MedImg	MRI	2k+	Brain tumor segmentation benchmark
CheXpert [60]	MedImg	X-ray	224k	Chest radiograph labels; uncertain labels handling
MIMIC-CXR [61]	MedImg	X-ray	370k	Large chest X-ray + reports; multi-modal learning
NIH ChestX-ray14 [62]	MedImg	X-ray	112k	Multi-label thoracic disease classification

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Dataset	Area	Modality	Scale	Typical tasks / notes
2.14 Fairness, robustness, and shift evaluation				
Waterbirds [63]	Robust	Images	~10k	Spurious correlation benchmark (background vs. label)
WILDS [64]	Robust	Mixed	10 datasets	In-the-wild distribution shifts across multiple application domains

3 Computer Vision

Computer vision benchmarks are commonly organized by task family, including image classification, object detection, segmentation, tracking, pose estimation, and three-dimensional (3D) perception. Whereas early benchmarks were generally modest in scale and tightly curated, contemporary CV research increasingly relies on large, heterogeneous datasets and standardized evaluation protocols, such as average precision for detection, intersection over union for segmentation, and task-specific leaderboard metrics. A further distinction separates datasets designed primarily for supervised evaluation from those used predominantly for representation learning or pretraining. More recently, multimodal vision–language datasets have become central to foundation-model research, further blurring the traditional boundary between CV and NLP and introducing additional concerns regarding web-scale data acquisition, annotation noise, provenance, duplication, and licensing.

Selecting an appropriate CV benchmark requires careful consideration of label taxonomies, annotation granularity—from image-level labels to instance masks and panoptic annotations—domain coverage, and the intended generalization regime. Many datasets contain systematic biases, including long-tailed class distributions, geographic or demographic imbalances, acquisition-specific artifacts, and contextual shortcuts, which can yield apparently strong but fragile improvements. Robustness-oriented benchmarks are designed to expose such failure modes through controlled distribution shifts, adversarial filtering, or transfer across domains such as natural images, artistic renditions, sketches, and stylized imagery. Meaningful comparison also depends on precise reporting of preprocessing procedures, input resolution, data augmentation, evaluation settings such as intersection-over-union thresholds or class averaging, and any external supervision or pretraining, because these choices can materially affect reported performance.

For concision, the accompanying table uses the following abbreviations:

- | | |
|--|--|
| • 2D = two-dimensional | • MedCV = medical computer vision |
| • 3D = three-dimensional | • MOT = multi-object tracking |
| • Ann = annotations | • MRI = magnetic resonance imaging |
| • CAD = computer-aided design | • OCR = optical character recognition |
| • Cls = classification | • Panoptic = panoptic segmentation |
| • CT = computed tomography | • Pose = two-dimensional pose estimation |
| • CV = computer vision | • Pose3D = three-dimensional pose estimation |
| • Det = object detection | • QA = question answering |
| • FaceAttr = face attribute recognition | • RGB = red, green, and blue |
| • FaceDet = face detection | • RGB-D = red, green, and blue plus depth |
| • FaceID = face identification or verification | • SemSeg = semantic segmentation |
| • Flow = optical flow | • SR = super-resolution |
| • GAN = generative adversarial network | • V+L = vision–language |
| • Gen = generative modeling | • VidSeg = video segmentation |
| • InstSeg = instance segmentation | • VQA = visual question answering |
| • LiDAR = light detection and ranging | |

Table 2: Computer vision datasets.

Dataset	Area	Modality	Scale	Typical tasks / notes
3.1 Image classification (general)				
CIFAR-10/100 [65]	Cls	Images	60k	Small natural images benchmark
Fashion-MNIST [66]	Cls	Images	70k	Apparel classification; MNIST-style
Food-101 [67]	Cls	Images	101k	Fine-grained food recognition with 101 categories
ImageNet [68]	Cls	Images	14M+	Large-scale classification; pretraining cornerstone
ImageNet-21k [69]	Cls	Images	14M+	Larger label space; used for pretraining
MNIST [70]	Cls	Images	70k	Handwritten digits; classic baseline
Places365 [71]	Cls	Images	1.8M+	Large-scale scene classification and representation learning
SVHN [72]	Cls	Images	600k	Street numbers; digit recognition
3.2 Fine-grained recognition				
CUB-200-2011 [73]	Fine-grained	Images	11k	Bird species classification with part annotations
FGVC Aircraft [74]	Fine-grained	Images	10k	Aircraft variant classification
iNaturalist [75]	Fine-grained	Images	Large	Long-tail species recognition
Oxford Flowers 102 [76]	Fine-grained	Images	8k+	Fine-grained flower species classification
Oxford-IIIT Pet [77]	Fine-grained	Images	7.3k	Pet breed classification and foreground segmentation
Stanford Cars [78]	Fine-grained	Images	16k	Car model classification
3.3 Object detection				
MS COCO [79]	Det	Images	330k	Detection + segmentation + keypoints
Objects365 [80]	Det	Images	2M	Large-scale object detection for diverse categories
Open Images [81]	Det	Images	9M+	Large-scale detection with many classes

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Dataset	Area	Modality	Scale	Typical tasks / notes
PASCAL VOC [82]	Det	Images	20 classes	Classic detection benchmark
3.4 Instance segmentation				
ADE20K (parts/instances) [83]	InstSeg	Images	25k+	Scene parsing with richer label structure
COCO (Instance Segmentation) [79]	InstSeg	Images	330k	Instance masks; common baseline
LVIS [84]	InstSeg	Images	1000+ classes	Long-tail instance segmentation
3.5 Semantic segmentation				
ADE20K [83]	SemSeg	Images	25k+	Scene parsing / dense semantic labels
Cityscapes [85]	SemSeg	Images	5k fine	Urban driving scenes; strong benchmark
COCO-Stuff [86]	SemSeg	Images	164k	Stuff + thing labels; semantic segmentation
Mapillary Vistas [87]	SemSeg	Images	25k	Street-scene semantic segmentation with fine-grained labels
PASCAL-Context [88]	SemSeg	Images	10k	Dense labels derived from PASCAL
3.6 Human pose estimation and keypoints				
COCO Keypoints [79]	Pose	Images	330k	Keypoint detection; standard evaluation
Human3.6M [89]	Pose3D	Video+3D	Large	3D pose and motion-capture benchmark
MPII Human Pose [90]	Pose	Images	25k+	2D pose-estimation benchmark
PoseTrack [91]	Pose	Video	Large	Pose tracking in videos
3.7 Face analysis				
CelebA [92]	FaceAttr	Images	200k	Face attributes and identities
FFHQ [93]	Faces	Images	70k	High-quality face images; generation and restoration
LFW [94]	FaceID	Images	13k	Face verification; legacy benchmark
WIDER FACE [95]	FaceDet	Images	32k	Face detection in the wild
3.8 Tracking and MOT				

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Dataset	Area	Modality	Scale	Typical tasks / notes
LaSOT [96]	Tracking	Video	1.4k	Single object long-term tracking
MOT17 [97]	Tracking	Video	~7 sequences	Multi-object tracking benchmark
MOT20 [98]	Tracking	Video	~4 sequences	Crowded multi-object tracking
3.9 Video action recognition				
HMDB51 [99]	Action	Video	7k	Action recognition; smaller benchmark
Kinetics [100]	Action	Video	Hundreds k	Human action recognition benchmark
Something- Something V2 [101]	Action	Video	220k	Fine-grained human-object interaction actions
UCF101 [102]	Action	Video	13k	Action recognition; classic dataset
3.10 Video segmentation				
DAVIS [103]	VidSeg	Video	Small	Video object segmentation benchmark
YouTube- VOS [104]	VidSeg	Video	Large	Large-scale video object segmentation
3.11 Optical flow and correspondence				
FlyingChairs [105]	Flow	Pairs	22k	Synthetic flow training dataset
FlyingThings3D [106]	Flow	3D	Large	Synthetic flow + disparity; training for correspondence
Middlebury Flow [107]	Flow	Pairs	Small	Classic optical flow evaluation
MPI-Sintel [108]	Flow	Video	Large	Synthetic optical flow benchmark with ground truth
3.12 Stereo and depth estimation				
KITTI [109]	Depth	RGB+LiDAR	Large	Stereo, flow, odometry, 3D detection
KITTI Depth [110]	Depth	RGB+LiDAR	Large	Depth completion/estimation tasks
NYU Depth v2 [111]	RGB-D	RGB-D	Small	Indoor RGB-D; depth + semantics
SUN RGB-D [112]	RGB-D	RGB-D	10k	Indoor scene understanding with depth
3.13 Autonomous driving (large-scale)				

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Dataset	Area	Modality	Scale	Typical tasks / notes
ApolloScape [113]	Drive	Images+3D	Large	Autonomous-driving scene understanding and localization
Argoverse [114]	Drive	Multi-sensor	Large	3D tracking and motion forecasting in urban traffic
BDD100K [115]	Drive	Video	100k	Driving video: detection, lane, segmentation
nuScenes [116]	Drive	Multi-sensor	1k scenes	360-degree multi-sensor perception benchmark
Waymo Open Dataset [117]	Drive	Multi-sensor	Large	3D detection and tracking using LiDAR and cameras
3.14 3D vision (scans, reconstruction, and point clouds)				
ModelNet40 [118]	3D	CAD	12k	3D shape classification baseline
S3DIS [119]	3D	Point cloud	Large	Indoor 3D semantic segmentation
ScanNet [120]	3D	RGB-D	Large	3D reconstruction + semantic labels
SemanticKITTI [121]	3D	LiDAR	43k scans	Semantic segmentation of sequential automotive LiDAR scans
ShapeNet [122]	3D	Meshes	Large	3D models for representation learning
3.15 Remote sensing and aerial imagery				
BigEarthNet [123]	Remote	Satellite	590k	Multi-label remote-sensing image classification
DeepGlobe [124]	Remote	Aerial	Large	Road/building segmentation challenges
DOTA [125]	Remote	Aerial	Large	Oriented object detection in aerial images
EuroSAT [126]	Remote	Satellite	27k	Land-use and land-cover classification from Sentinel-2 imagery
xView [127]	Remote	Aerial	1M objects	Large-scale overhead object detection
3.16 Document AI / OCR				
DocVQA [128]	Document AI	Images+QA	50k QA pairs	Visual question answering over document images
IAM Handwriting [129]	Document AI	Scans	Large	Handwriting recognition
PubTabNet [130]	Document AI	Scans	500k	Table structure recognition

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Dataset	Area	Modality	Scale	Typical tasks / notes
3.17 Vision–language (image–text)				
CLEVR [131]	V+L	Images+QA	100k	Synthetic visual reasoning and compositional question answering
Conceptual Captions (CC12M) [132]	V+L	Image-Text	12M	Larger caption dataset
Conceptual Captions (CC3M) [133]	V+L	Image-Text	3M	Caption pairs for multimodal pre-training
Flickr30k [134]	V+L	Image-Text	31k	Image captioning + retrieval
GQA [135]	V+L	Images+QA	22M QA pairs	Compositional visual reasoning on real images
LAION-5B [136]	V+L	Image-Text	5B	Web-scale image-text pairs; foundation-model pretraining
MSR-VTT [137]	V+L	Video-Text	10k	Video captioning / retrieval
Visual Genome [138]	V+L	Images+Ann	108k	Objects/attributes/relations; region descriptions
VQA [139]	V+L	Images+QA	Large	Visual question answering benchmark
3.18 Generative modeling and super-resolution				
CelebA-HQ [140]	Gen	Images	30k	High-quality faces for GANs
DIV2K [141]	SR	Images	2k	Super-resolution benchmark
FFHQ [93]	Gen	Images	70k	High-quality face dataset for generation and editing
3.19 Robustness and distribution shift				
ImageNet-A [142]	Robust	Images	7.5k	Adversarially filtered natural images
ImageNet-R [143]	Robust	Images	30k	Renditions (art, cartoons); domain shift
ImageNet-Sketch [144]	Robust	Images	50k	Sketch domain shift evaluation
Stylized-ImageNet [145]	Robust	Images	Large	Style transfer augmentation; texture/shape bias studies
3.20 Medical imaging (CV-specific)				

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Dataset	Area	Modality	Scale	Typical tasks / notes
ACDC [146]	MedCV	Cardiac MRI	150	Cardiac structure segmentation and diagnosis
BraTS [59]	MedCV	MRI	2k+	Brain tumor segmentation benchmark
DRIVE [147]	MedCV	Fundus	40	Retinal vessel segmentation (classic)
ISIC 2018 [148]	MedCV	Dermoscopic	10k+	Skin-lesion analysis and melanoma classification
Kvasir-SEG [149]	MedCV	Endoscopy	1k	Polyp segmentation in gastrointestinal endoscopy
LUNA16 [150]	MedCV	CT	888	Lung nodule detection benchmark

4 Conclusion and Outlook

This manuscript consolidates a representative collection of datasets and benchmarks for ML and CV, emphasizing practical descriptors such as modality, approximate scale, and typical tasks, together with benchmark hubs that standardize data access and evaluation. By providing a structured map of commonly used resources across areas ranging from tabular learning and time-series analysis to vision–language modeling and medical imaging, the overview can support benchmark selection, baseline development, comparative experimentation, and transparent reporting. The tables should nevertheless be regarded as informed entry points rather than prescriptive recommendations. The suitability of a benchmark ultimately depends on the scientific question, intended deployment context, available supervision, target population or domain, and required generalization setting.

Several developments are likely to shape future benchmarking practice. Evaluation will increasingly extend beyond a single predictive metric to encompass complementary dimensions such as calibration, fairness, robustness, computational efficiency, and energy or monetary cost. For multimodal and foundation models, credible evaluation will require clearer separation of pretraining sources from downstream benchmarks, accompanied by systematic assessment of data contamination, overlap, and deduplication. Dataset documentation and governance—including licensing, consent, provenance, and privacy constraints—will also become increasingly consequential, particularly for web-scale and healthcare datasets. At the same time, continuously maintained repositories and community leaderboards will continue to evolve, making explicit dataset versioning and reproducible evaluation pipelines indispensable for reliable comparison. Accordingly, this overview is intended to remain updateable while promoting benchmark selection and reporting that are transparent, reproducible, and scientifically justified.

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